

# 1. Purpose, Scope and Safety Boundary

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

## Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

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## 2. What Is Aircraft Fuel Leakage?

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| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
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### 3. Why Fuel Leakage Matters in Aircraft Manufacturing

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| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

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## 4. Generic Fuel-System Leakage Locations

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| Control layer | Question                                          | Example evidence        |
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| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
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## 5. Types of Fuel Leakage

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| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
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## 6. The Fuel Leakage Chain

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| Data field     | Why it matters              | Example      |
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## 7. Leakage Indicators

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| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
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## 8. Step-by-Step Leakage Investigation

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## 9. Fuel Quantity Difference Calculation

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For a project demonstration, calculations are used to make reasoning explicit. Illustrative numbers below are examples only; they are not aircraft acceptance criteria. The software should display the formula, inputs, units and result so the reviewer can verify the arithmetic.

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### Illustrative quantity-difference calculation

Expected quantity = 120.0 L; measured quantity = 117.0 L. Therefore,  $\Delta V = 120.0 - 117.0 = 3.0$  L. Using an illustrative classroom density of 0.80 kg/L,  $\Delta m = 3.0 \times 0.80 = 2.4$  kg. The result is only a screening difference; it does not prove leakage.

| Data field     | Why it matters              | Example      |
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## 10. Leak Rate Calculation

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### Illustrative leak-rate calculation

If a measured change is 1.5 L over 30 min,  $Q = \Delta V / \Delta t = 1.5 / 30 = 0.05$  L/min. Store the start time, end time, quantities, units and uncertainty. Never use this generic number as an aircraft-specific limit.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
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# 11. Measurement Uncertainty

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## Illustrative uncertainty comparison

Observed difference = 3.0 L; illustrative combined uncertainty =  $\pm 1.0$  L. Because 3.0 L is greater than the example uncertainty magnitude, the case is a stronger signal for investigation. If the difference were 0.6 L, the evidence would be less convincing. Real uncertainty must come from the approved method.

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## 12. Temperature Effects on Fuel Quantity

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## 13. Potential Causes of Fuel Leakage

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| Control layer | Question                                          | Example evidence        |
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## 14. FMEA for Fuel Leakage

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## 15. Illustrative Risk Calculation

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### Illustrative project risk score

Project Risk Screen = Severity x Likelihood. If Severity = 5 and Likelihood = 2, the score is 10. If Severity = 3 and Likelihood = 4, the score is 12. These are only internal prioritisation examples, not airworthiness classifications.

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## 16. Leakage Detection Methods

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A good prevention method is layered: prevent the error where possible, detect deviations early, preserve evidence, and feed confirmed causes back into the process. A method is valuable when it is repeatable, explainable, traceable and compatible with the organisation's approved workflow.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.



## 17. Visual Inspection and Evidence Quality

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 18. Fuel Leakage During Manufacturing and Assembly

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 19. Maintenance-Related Leakage - Learning Logic

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 20. Leakage After Refuelling

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

Fuel handling combines liquid fuel, vapour and possible ignition sources. Current EASA Air Operations rules include specific precautions for refuelling and state that refuelling should be stopped when fuel vapour is detected inside the aircraft or another hazard arises in the covered operation. Actual site procedures always take precedence.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

**Refuelling safety note: EASA's current Air Operations material requires risk controls around special refuelling/defuelling and states, in covered procedures, that if fuel vapour is detected inside the aircraft or another hazard arises during refuelling, refuelling should be stopped immediately. This is a general regulatory reference, not a HAL-specific procedure.**

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 21. Water in Aviation Fuel

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

**FAA AC 20-125 is an active advisory circular on water in aviation fuels. It explains that water can enter fuel systems through pathways including leaks in vents or seals, poorly fitting filler interfaces, refuelling equipment, condensation and precipitation, and recommends prevention/detection/elimination controls.**

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 22. Fuel Contamination and Leakage - Connected Risks

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 23. Leak Path and Flow Concept

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 24. Seals and Interfaces

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.



## 25. Lines, Hoses and Fittings

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 26. Tanks and Access Interfaces

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 27. Fuel Vapour and Leakage

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

Fuel handling combines liquid fuel, vapour and possible ignition sources. Current EASA Air Operations rules include specific precautions for refuelling and state that refuelling should be stopped when fuel vapour is detected inside the aircraft or another hazard arises in the covered operation. Actual site procedures always take precedence.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 28. Fire, Static and Ignition Awareness

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

Fuel handling combines liquid fuel, vapour and possible ignition sources. Current EASA Air Operations rules include specific precautions for refuelling and state that refuelling should be stopped when fuel vapour is detected inside the aircraft or another hazard arises in the covered operation. Actual site procedures always take precedence.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 29. Human Factors

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 30. Root Cause Analysis - 5 Why

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 31. Fishbone Analysis

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 32. Pareto Analysis

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

### Illustrative Pareto calculation

Suppose 100 records contain 35 fitting/interface cases, 25 line/hose, 20 seal, 10 valve/equipment and 10 unknown. Cumulative shares are 35%, 60%, 80%, 90% and 100%. The first three groups account for 80% of records in this example.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.



## 33. Trend Analysis

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

### Illustrative normalised trend

Month 1: 8 findings / 20 units = 0.40 per unit. Month 6: 3 findings / 50 units = 0.06 per unit. The denominator matters because raw defect counts can be misleading when production volume changes.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 34. Leakage Cost Calculation

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

For a project demonstration, calculations are used to make reasoning explicit. Illustrative numbers below are examples only; they are not aircraft acceptance criteria. The software should display the formula, inputs, units and result so the reviewer can verify the arithmetic.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

### Illustrative cost calculation

Rework = 6 h x INR 500/h = INR 3,000. Inspection = 2 h x INR 500/h = INR 1,000. Consumables = INR 800. Direct illustrative total = INR 4,800. Real organisation-specific rates must come from authorised data.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 35. Detection Coverage Calculation

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

For a project demonstration, calculations are used to make reasoning explicit. Illustrative numbers below are examples only; they are not aircraft acceptance criteria. The software should display the formula, inputs, units and result so the reviewer can verify the arithmetic.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

### Illustrative detection metrics

For 1,000 synthetic leakage cases, a detector that correctly flags 940 gives detection rate =  $940/1000 \times 100 = 94\%$ . If 8 of 100 normal cases are incorrectly flagged, the false-positive rate is 8%. These values describe the test set only.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 36. 10,000-Case Software Testing

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

A large synthetic test set is useful for checking software logic. Every test case should have an expected answer generated by an independent rule or oracle. Passing 10,000 synthetic cases proves consistency against those rules; it does not certify a system for real aircraft use.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

### 10,000-case test design

A practical split is 2,000 normal cases, 2,000 clear leakage cases, 1,500 measurement-noise cases, 1,500 missing-data cases, 1,000 conflicting-record cases, 1,000 boundary cases and 1,000 random-combination cases. Total = 10,000 cases. Each case needs an expected result.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 37. Offline Fuel-Leakage Analysis Tool Concept

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The software concept is intentionally offline. It can run on a local computer using synthetic or approved local data, calculate screening indicators, identify missing information, compare recurring patterns and generate a local report. It should not make an aircraft-release decision.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 38. Software Data Model

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The software concept is intentionally offline. It can run on a local computer using synthetic or approved local data, calculate screening indicators, identify missing information, compare recurring patterns and generate a local report. It should not make an aircraft-release decision.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 39. Validation Rules

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 40. Single End-to-End Workflow

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.



## 41. Example Case - Quantity Difference

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 42. Example Case - Visible Wetness

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 43. Example Case - Repeated Pattern

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 44. Prevention - Design and Process

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

A good prevention method is layered: prevent the error where possible, detect deviations early, preserve evidence, and feed confirmed causes back into the process. A method is valuable when it is repeatable, explainable, traceable and compatible with the organisation's approved workflow.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 45. Prevention - Traceability

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

A good prevention method is layered: prevent the error where possible, detect deviations early, preserve evidence, and feed confirmed causes back into the process. A method is valuable when it is repeatable, explainable, traceable and compatible with the organisation's approved workflow.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 46. How to Judge Whether a Method Is Good

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

Safety boundary: this student reference does not provide aircraft-specific torque values, leak limits, pressure values, inspection intervals, flight-crew procedures, dispatch rules, or repair instructions. Actual aircraft work must use the applicable approved technical publications and authorised engineering, maintenance, quality and airworthiness personnel.

## 47. Limitations and Non-Assumptions

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

| Observation       | Meaning                                 | Caution                              |
|-------------------|-----------------------------------------|--------------------------------------|
| Visible wetness   | Fuel is present at a location           | Source may be elsewhere              |
| Quantity mismatch | Measured quantity differs from expected | Measurement error must be considered |
| Odour             | Possible vapour/fuel presence           | Subjective; not proof by itself      |
| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

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## 48. Future Intelligent Offline Prevention

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A good prevention method is layered: prevent the error where possible, detect deviations early, preserve evidence, and feed confirmed causes back into the process. A method is valuable when it is repeatable, explainable, traceable and compatible with the organisation's approved workflow.

### Step-by-step method

- 1. Observe:** Write exactly what was seen or measured; do not turn an observation into a root cause.
- 2. Record:** Capture time, location, configuration, measurement, instrument and evidence identifiers.
- 3. Validate:** Check units, timing, duplicate records and measurement status.
- 4. Compare:** Compare expected and measured data only when the measurement conditions are comparable.
- 5. Investigate:** Map possible sources and supporting evidence.
- 6. Verify:** Use the approved technical method for any actual aircraft confirmation.
- 7. Learn:** Convert the final cause into a prevention or process-improvement lesson.

### Future decision logic

A future intelligent version can combine deterministic rules with anomaly detection. The rule engine can answer known conditions; a machine-learning model can flag unusual patterns; the engineer can review the evidence. This human-in-the-loop design is safer than allowing the model to make an aircraft release decision.

| Data field     | Why it matters              | Example      |
|----------------|-----------------------------|--------------|
| Case ID        | Unique traceability         | FL-2026-0042 |
| Location code  | Grouping and source mapping | F-03         |
| Timestamp      | Sequence and correlation    | 14:10        |
| Measured value | Calculation input           | 117.0 L      |
| Uncertainty    | Confidence context          | +/- 1.0 L    |
| Evidence ID    | Supports conclusion         | E17          |

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## 49. 20-Day Internship Implementation Plan

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

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| Control layer | Question                                          | Example evidence        |
|---------------|---------------------------------------------------|-------------------------|
| Design        | Can the boundary perform its intended function?   | Approved design data    |
| Process       | Was the assembly/manufacturing method controlled? | Work record             |
| Inspection    | Was the result checked?                           | Inspection/test result  |
| Traceability  | Can the finding be linked back to the item?       | Case/part/evidence IDs  |
| Feedback      | Was recurrence analysed?                          | Trend/root-cause record |

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## 50. Final Summary and Reference Sources

Aircraft fuel leakage should be treated as a controlled engineering and safety problem. The objective of this manual is to explain the topic in simple language while keeping the engineering reasoning traceable. A leak can be visible, intermittent, hidden, quantity-based, vapour-related, or discovered indirectly through records. The correct approach is to preserve evidence, separate observation from hypothesis, use approved technical information, and document the reasoning so another reviewer can reproduce it.

The engineering goal is to replace vague statements such as 'fuel leak found' with structured evidence. A clear record identifies what was observed, where, when, under what condition, how it was measured, what is known, what is still uncertain, and what approved reference governs the next action.

### Step-by-step method

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| Observation       | Meaning                                 | Caution                              |
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| Residue/stain     | Historical indication may remain        | May not identify timing              |
| Sensor anomaly    | Measured signal outside pattern         | Sensor health matters                |

**Reference sources:** HAL Aircraft Division Bangalore - public production-unit overview; FAA AC 20-125 - Water in Aviation Fuels; FAA AC 33.67-1 - Engine Fuel Systems; EASA Easy Access Rules for Air Operations, Revision 24, March 2026. These public sources provide general context. Controlled HAL documents and aircraft-type manuals take precedence for actual work.

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